Abhiram Mullapudi

abhiramm@umich.edu abhiramm7 randomstorms.net

Experience

Lead Data Scientist, *Inframark* 2025–Present
Leading the development of machine learning-based solutions for optimizing the operation of collection systems and wastewater treatment plants.

Senior Data Scientist, *Xylem* 2023–2025
Designed and implemented end-to-end machine learning-based solutions that inform decision-making in urban water infrastructure systems.

- Led the development of statistical and machine learning-based methodologies for time-series filtering and anomaly detection for predictive maintenance and operational decision-making.
- Developed a Flyte-based MLOps platform to streamline end-to-end machine learning model development, deployment, and maintenance for Xylem's digital water products.

Hydraulic Control and Optimization Engineer, *Xylem* 2020–2023
Pioneered advanced machine learning and data engineering solutions for urban water infrastructure, transforming raw sensor data into actionable intelligence that optimizes water network performance.

- Developed a 1D-CNN model that leverages NOAA rainfall forecasts and near-real-time flow measurements to accurately predict 24-hour inflow to water treatment plants.
- Engineered an advanced 1D-CNN interpolation framework for processing spatially distributed river level data, enabling comprehensive environmental monitoring.
- Designed a high-performance real-time processing system leveraging symbolic programming and advanced statistical techniques to detect network irregularities across 600+ concurrent data streams.

Publications

Identification of stormwater control strategies and their associated uncertainties using Bayesian Optimization 2023
arXiv preprint

pystorms: a simulation sandbox for the design and evaluation of stormwater control algorithms 2023
Environmental Modelling and Software

Improvement of phosphorus removal in bioretention cells using real-time control 2022
Urban Water Journal

StormReactor: An open-source Python package for the integrated modeling of urban water quality and water balance 2021
Environmental Modelling & Software

Deep Reinforcement Learning for the Real Time Control of Stormwater Systems 2020
Advances in Water Resources

PySWMM: The Python Interface to Stormwater Management Model (SWMM) 2020
Journal of Open Source Software

rrcf: Implementation of the Robust Random Cut Forest algorithm for anomaly detection on streams 2019
The Journal of Open Source Software

Shaping Streamflow Using a Real-Time Stormwater Control Network 2018
Sensors

Emerging investigators series: building a theory for smart stormwater systems 2017
Environmental Science: Water Research & Technology

Education

Ph.D. in Civil Engineering
2017–2020 University of Michigan
Statistical Learning Approaches for the Control of Stormwater Systems
Advisor: Dr. Branko Kerkez

M.Sc.Eng. in Civil Engineering
2015–2017 University of Michigan

B.Tech. in Civil Engineering
2011–2015 Amrita Vishwa Vidyapeetham

Skills

Water Systems Modeling
EPA-SWMM, pyswmm, pystorms, digital twins, hydraulic & hydrologic modeling

ML & Optimization
PyTorch, TensorFlow, JAX, scikit-learn; time-series forecasting, anomaly detection, reinforcement learning, Bayesian optimization, physics-informed ML

Sensing & Embedded Systems
Wireless sensor networks, Open-Storm `perfect-cell`, EAGLE, Cypress

Programming
Python, MATLAB, C/C++, SQL, Bash

Cloud & MLOps
Flyte, MLflow, AWS, Google Cloud, Azure

Awards

LIFT Intelligent Water Systems Challenge
Grand Prize Winner, 2019

Academic Excellence
Amrita Vishwa Vidhyapeetham, 2013, 2015

Projects & Software

pystorms Creator
Python library for the design and evaluation of stormwater control algorithms.

pyswmm Maintainer
The Python interface to EPA-SWMM, the industry standard for stormwater modeling.

StormReactor Co-creator
Integrated water-quality and water-balance modeling for EPA-SWMM.

OpenFloodHub Creator
Open flood forecasting: a per-gauge CNN predicts 12-hour streamflow, benchmarked against NOAA's National Water Model.

flood sim Creator
A GPU shallow-water flood model that runs in the browser, built from open terrain and streamflow data.